U.S. Retail Sales Trends

A Data Analysis & Visualization Project (1992–2026)

This report analyzes 34+ years of U.S. Advance Retail Sales data (Retail Trade and Food Services), sourced from the U.S. Census Bureau via the Federal Reserve Economic Data (FRED) platform. It covers overall growth, recession impacts, seasonal patterns, and a 12-month forward trend forecast.

Metric	Value
Data range	1992-01-01 – 2026-07-01
Observations	415 monthly data points
Latest value	$763,602M (July 2026)
Latest YoY growth	5.0%
Total growth over period	380%
COVID-19 shock (Feb→Apr 2020)	-22.2%
2008 Financial Crisis (Jul→Dec 2008)	-11.7%
12-month forecast (by Jul 2027)	$775,197M

Source: U.S. Census Bureau, Advance Retail Sales: Retail Trade and Food Services [RSAFS], retrieved from FRED, Federal Reserve Bank of St. Louis (fred.stlouisfed.org/series/RSAFS). Units: Millions of Dollars, Seasonally Adjusted.

1. Long-Run Trend

Since January 1992, U.S. retail sales have grown from $159,177M to $763,602M per month as of July 2026 — a cumulative increase of 380%. The series shows a persistent long-run upward trend, interrupted by two sharp downturns: the 2008 Global Financial Crisis and the 2020 COVID-19 pandemic. Both are visible as clear dips below the trend line before the series resumes its climb.

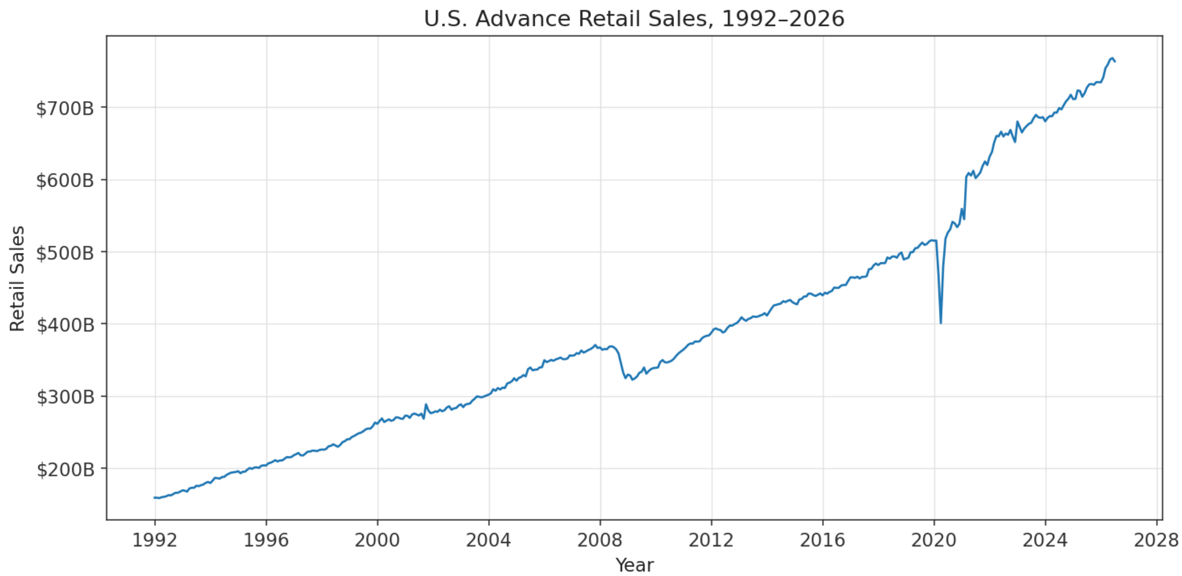

Figure 1. Monthly U.S. retail sales, 1992–2026 (seasonally adjusted, millions of dollars).

2. Growth Rate & Recession Impact

Year-over-year (YoY) growth is typically in the 3–8% range during normal expansions. The shaded bands below mark three NBER-dated U.S. recessions that fall within the sample: the 2001 downturn, the 2008–2009 Global Financial Crisis, and the brief but severe 2020 COVID-19 recession. Growth turned negative during each. The 2008 crisis produced the deepest sustained decline (sales fell 11.7% from their July 2008 peak to the December 2008 trough), while COVID caused the single sharpest monthly drop (22.2% from February to April 2020) but also the fastest recovery, aided by stimulus spending and a rapid shift to e-commerce.

Figure 2. Year-over-year % change, with recession periods shaded grey.

3. Smoothing Out Month-to-Month Noise

Raw monthly sales are noisy — driven by holidays, weather, and one-off events. A 12-month rolling average removes this short-term noise and makes the underlying trajectory much easier to read, while month-over-month (MoM) % change highlights the volatility itself, including the extraordinary spikes and crashes of 2020.

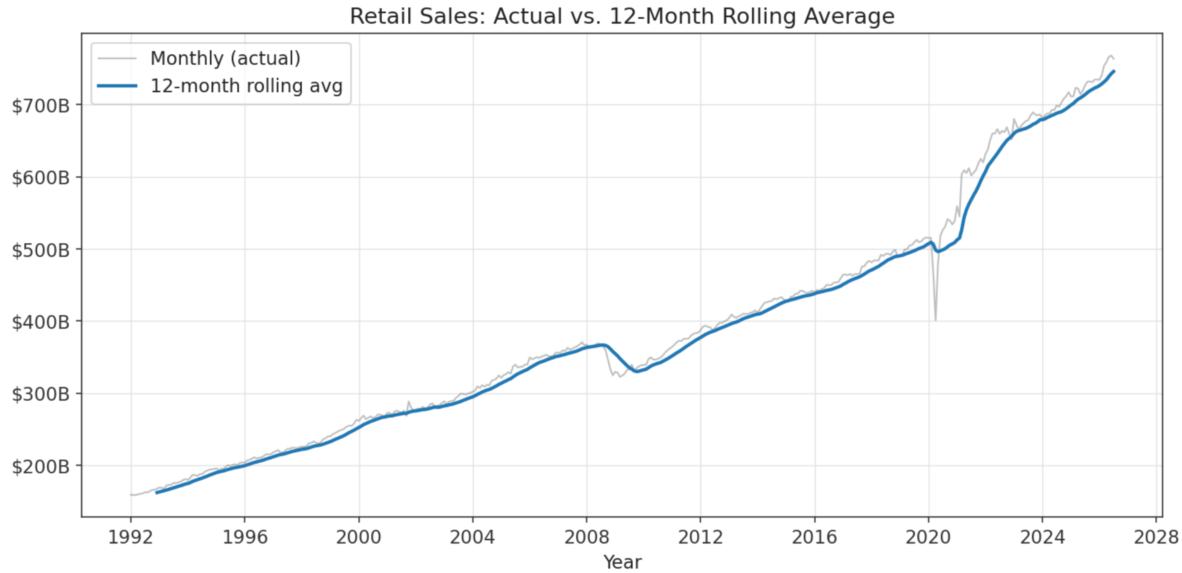

Figure 3. Actual monthly sales vs. 12-month rolling average.

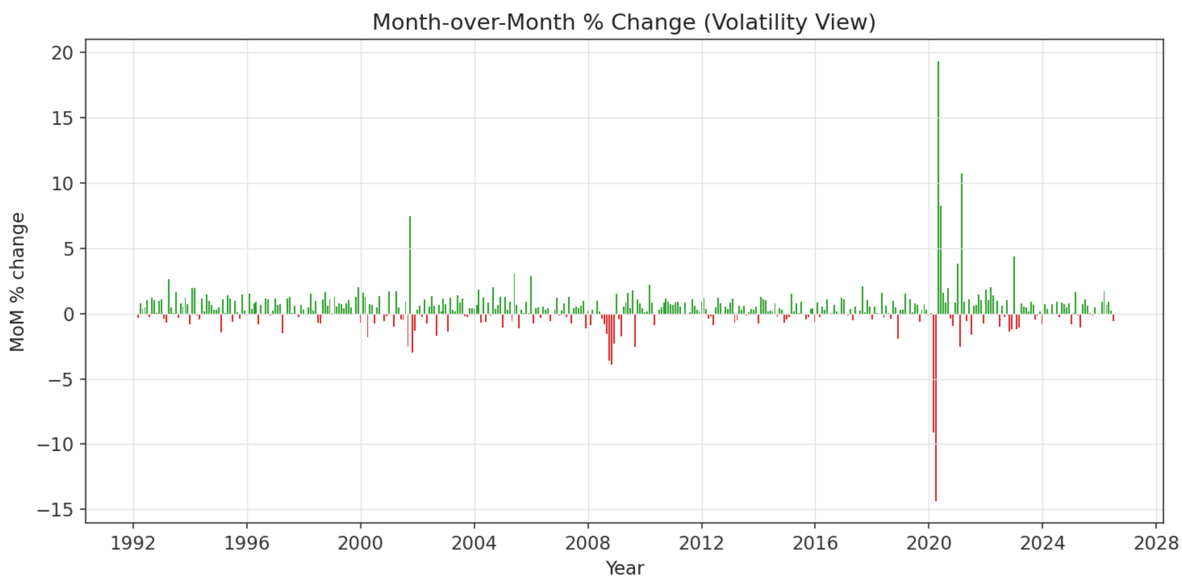

Figure 4. Month-over-month % change — note the 2020 whipsaw.

4. Case Study: The COVID-19 Shock

Retail sales fell from $515,330M in February 2020 to just $401,028M in April 2020 — a 22.2% drop in two months, the fastest collapse in the dataset. The recovery was equally fast: sales surpassed their pre-pandemic level by June 2020 and then kept climbing well above trend through 2021, fueled by stimulus payments, pent-up demand, and elevated goods spending while services were still constrained.

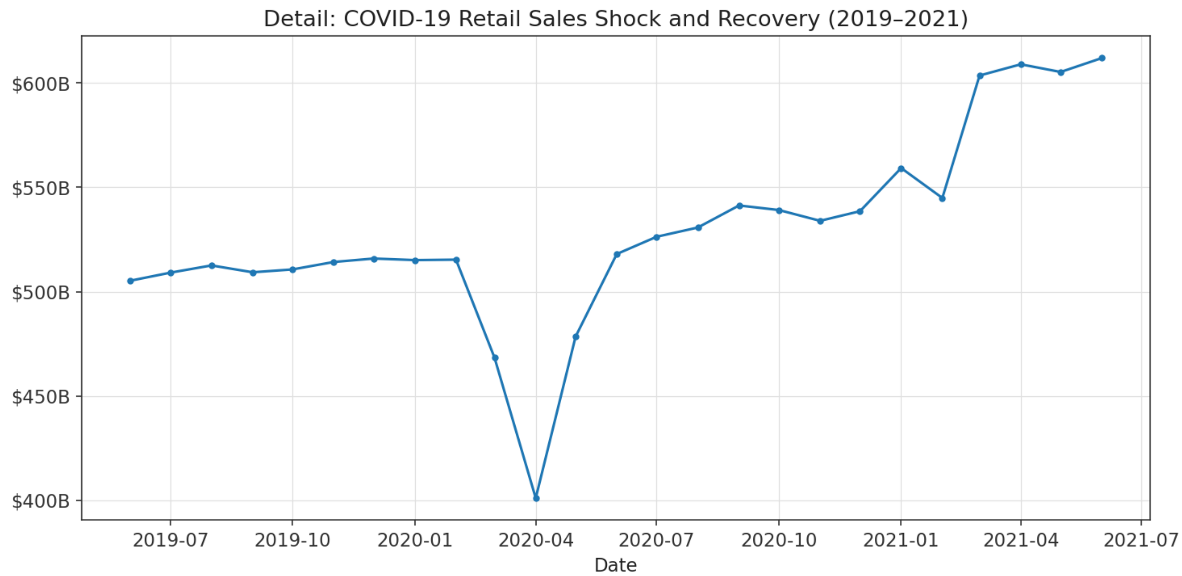

Figure 5. Detail view of the COVID-19 shock and recovery, 2019–2021.

5. Methodology: Seasonal Decomposition & Trend Model

To separate the underlying trend from seasonal and irregular effects, this project applies a classical multiplicative decomposition, implemented from scratch in `model/trend_model.py` (no black-box forecasting library required):

• **Trend-cycle:** a 2×12 centered moving average, which removes both seasonal and short-term noise, leaving the smooth long-run trajectory.
• **Seasonal index:** the average ratio of actual sales to trend for each calendar month, normalized to average 1.00 across the year.
• **Remainder:** whatever is left after dividing out trend and seasonality — ideally small, random noise.

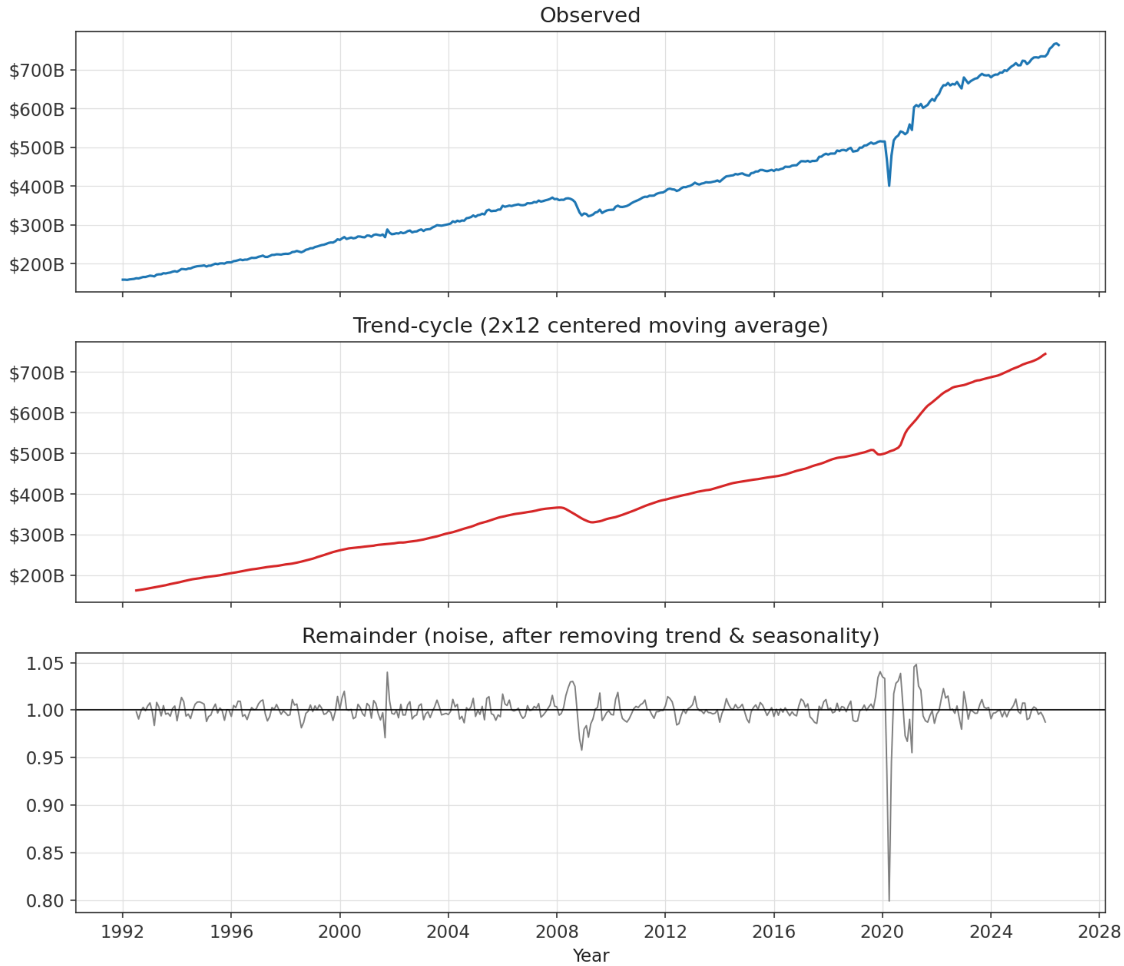

Figure 6. Observed series decomposed into trend-cycle and remainder.

5.1 A Note on Seasonality in This Series

The Census Bureau publishes RSAFS already **seasonally adjusted**. Running our own seasonal decomposition on top of an already-adjusted series is expected to recover a nearly flat seasonal index — which is exactly what we find (all monthly indices fall within about ±0.5% of 1.00). This is a useful sanity check: it confirms the Census Bureau's adjustment is doing its job, rather than indicating a bug in the analysis.

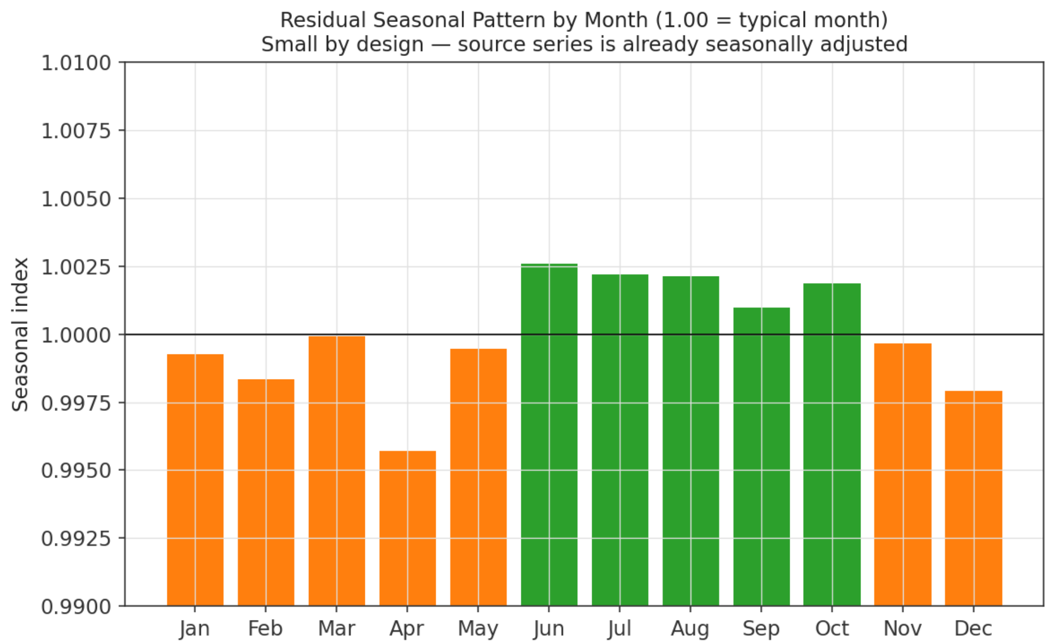

Figure 7. Residual seasonal pattern, zoomed in — small by construction.

5.2 Forecasting Approach

A straight line fit to the *entire* 1992–2026 trend-cycle explains 93% of its variance (R-squared = 0.93) and implies average growth of about $15,239M/year. But retail sales growth has clearly accelerated since 2021, so a single long-run line understates the current pace. The forecast instead uses a **local linear trend** fit on just the most recent 60 months (R-squared = 0.95), implying a current growth rate of roughly $29,096M/year — nearly double the long-run average. This local trend is projected forward 12 months and re-seasonalized using the monthly seasonal index above.

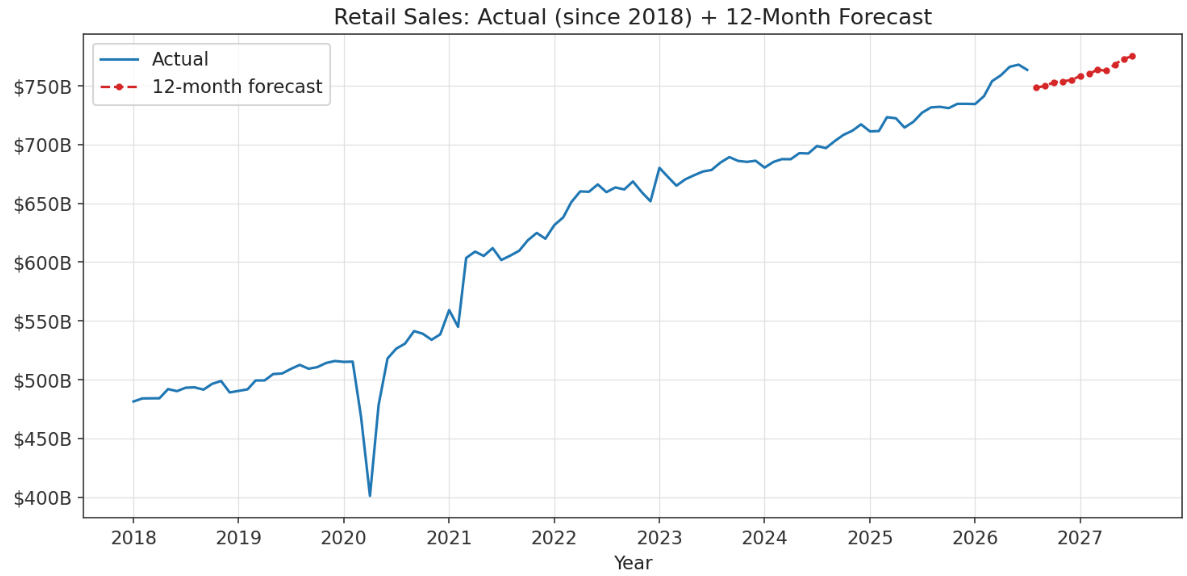

Figure 8. Actual sales since 2018 plus the 12-month forecast, reaching $775,197M by July 2027.

5.3 Limitations

This is a simple, transparent trend model — not a production forecasting system. It assumes recent growth continues linearly, does not model macroeconomic shocks, inflation, interest rates, or consumer sentiment, and would not have predicted the 2008 or 2020 downturns before they happened. Nominal dollars are used throughout (not inflation-adjusted); a meaningful share of the 1992–2026 growth reflects price inflation rather than real volume growth. Treat the forecast as a trend extrapolation, not a prediction.

6. Data & Reproducibility

All data, code, and this report are included in the accompanying repository so the full pipeline can be re-run end-to-end.

File	Purpose
data/retail_sales_raw.csv	Raw monthly series as retrieved from FRED (RSAFS)
data/process_data.py	Feature engineering: MoM/YoY %, rolling stats, recession flags
data/retail_sales_processed.csv	Output of process_data.py — analysis-ready dataset
model/trend_model.py	Seasonal decomposition + local linear trend forecast model
data/retail_sales_forecast.csv	12-month forecast output
model/model_summary.txt	Plain-text model diagnostics (R-squared, growth rates, forecast table)
analysis.py	Generates every chart in this report
charts/*.png	All figures used in this report
report/generate_report.py	Builds this PDF report

To reproduce:

```
pip install -r requirements.txt
python data/process_data.py
python model/trend_model.py
python analysis.py
python report/generate_report.py
```

7. Citation

U.S. Census Bureau, Advance Retail Sales: Retail Trade and Food Services [RSAFS], retrieved from FRED, Federal Reserve Bank of St. Louis; https://fred.stlouisfed.org/series/RSAFS.